

Supersymmetric QFTs, CFTs and Integral Modular Functions

Introduction to elliptic genera

Nivedita, Indian Institute of Technology, Kanpur.



Introduction in Conformal Field Theory

Motivation

- We wish to find deformation invariants of QFTs, i.e. quantities that do not change on perturbation, or going along RG flows.
- Thus, these quantities will be equal for a QFT and the CFT that it flows to under RG.
- These quantities can help us understand the path-components of the space of QFTs using the CFT they flow to. [GJFW19]
- Reversely, we can try to quantify the space of QFTs geometrically and get a richer knowledge about the structure of the subspace containing CFTs. This will help in the classification of phases in condensed matter. [GJF19]
- We will work in the space of $\mathcal{N} = (\mathbf{0}, \mathbf{1})$ supersymmetric QFTs as such deformation invariants are easier to find in this case.

Graded vector spaces

In a boundary QFT, we evaluate our QFT not only on closed manifolds, but also on manifolds with boundary. The boundaries are assigned a Hilbert spaces while the manifold itself is assigned a linear map between these Hilbert spaces.

In the supersymmetric case, our Hilbert spaces are $\mathbb{Z}_2$ -graded (they can be written as $\mathcal{H}_0 \oplus \mathcal{H}_1$). We define the supersymmetric braiding of the tensor product of two super vectorspaces $\mathbf{V}, \mathbf{W}$ by

$$\begin{aligned}\tau: \mathbf{V} \otimes \mathbf{W} &\rightarrow \mathbf{W} \otimes \mathbf{V} \\ \mathbf{v} \otimes \mathbf{w} &\mapsto (-1)^{|\mathbf{v}||\mathbf{w}|} \mathbf{w} \otimes \mathbf{v}\end{aligned}$$

for elements of odd or even degree, and extend it by linearity.

We call an operator **even** if it preserves the grading and **odd** if it reverses it.

Super Lie algebras

Any operator is the sum of an even and odd operator. This gives rise to the supercommutator of operators on super vector spaces

$$[A, B] = AB - (-1)^{|A||B|} BA$$

for even or odd operators. Extend to all operators by linearity.

A super Lie algebra is a super vector space with a bilinear map $[-, -] : \mathbf{V} \otimes \mathbf{V} \rightarrow \mathbf{V}$ such that

$$[A, B] = -(-1)^{|A||B|} [B, A]$$

for homogenous elements of the super vector space. The bracket must also satisfy a version of the Jacobi identity.

The above definition of the supercommutator of operators turns the space of operators into a super Lie algebra.

Supertrace

Supertrace of any operator is defined as

$$\text{str } \mathcal{O} = \text{tr } \mathcal{O}_0 - \text{tr } \mathcal{O}_1$$

where **0** and **1** are the even and odd components of the operator respectively.

Also, (super)trace of a (super)commutator vanishes. (take cases and evaluate)

Supersymmetry

By (Poincaré) supersymmetry we mean a super Lie algebra whose even part is the Poincaré algebra. It will have at least one of two super Poincaré algebras called the $\mathcal{N} = (0, 1)$ and $\mathcal{N} = (1, 0)$ algebras as its subalgebras.

We say that a 2 dimensional QFT has $\mathcal{N} = (0, 1)$ supersymmetry when the fields defining the theory form a representation of the $\mathcal{N} = (0, 1)$ super-Poincaré algebra.

This is the usual Poincaré algebra equipped with an extra element Q , satisfying

$$\{Q, Q\} = P_y - iP_x = -2i\bar{P}$$

and Q commutes with P_i, L

Background

When we want to study QFTs invariant under Poincaré transformations, a natural way to do it is study fields defined on $\mathbb{R}^n$ such that the action only involves invariant terms.

We wish to find a similar geometric space for the above supersymmetric algebra. We will call this $\mathbb{R}^{n|m}$.

Superspace is an extension of ordinary spacetime to include extra anticommuting coordinates.

cs manifolds

A complex super (cs) manifold of dimension $n|m$ is a space such that locally, the $\mathbb{C}$ -valued scalar fields on it look like

$$\mathcal{C}^\infty(\mathbb{R}^n, \mathbb{C}) \otimes \Lambda[\theta_1, \dots, \theta_m]$$

where $\Lambda[\theta_1, \dots, \theta_m]$ is the algebra over $\mathbb{C}$ generated by m elements satisfying

$$\theta_i \theta_j = -\theta_j \theta_i$$

Define the space $\mathbb{R}^{2|1}$ to be parametrised by three coordinates $(z, \bar{z}, \theta)$ such that $z, \bar{z}$ are complex and θ is Grassmann. This is obviously a cs manifold of dimension $2|1$. We give it a super Lie group structure by defining multiplication by

$$(z_1, \bar{z}_1, \theta_1) +_s (z_2, \bar{z}_2, \theta_2) = (z_1 + z_2, \bar{z}_1 + \bar{z}_2 + \theta_1 \theta_2, \theta_1 + \theta_2)$$

This is the exponentiation of the $\mathcal{N} = (0, 1)$ super Poincare algebra.

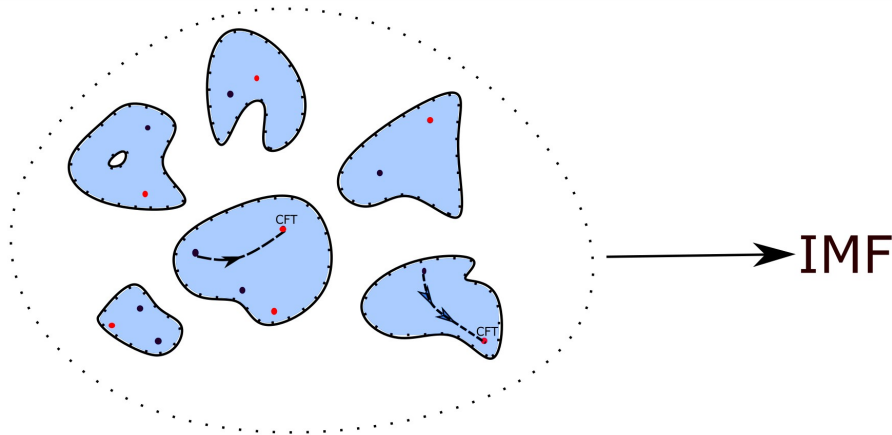
Main theorem [ST11]

The partition function of a 2d $\mathcal{N} = (0, 1)$ supersymmetric QFT evaluated on a torus is an integral modular function. Further, this map is surjective. This is a deformation invariant.

We will only prove the first part of the theorem.

- An integral modular function is a meromorphic function $f: \mathbb{H} \rightarrow \mathbb{C}$ invariant under the $\mathrm{SL}_2(\mathbb{Z})$ action and has integral Fourier coefficients

Result



Theorem 1

Some Geometric Facts

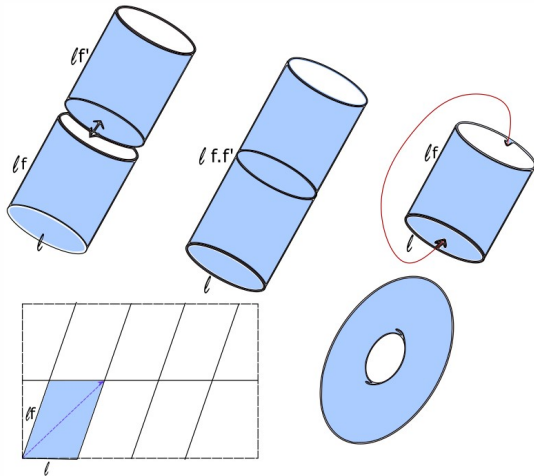


Figure: $C_{I,f}$ is a super-cylinder i.e. its two boundaries are S_I (super circle with circumference I). The cylinder is quotient of $\mathbb{R}^2$ by a lattice, with parameters I and If

Geometric facts

- $C_{I,f} \circ C_{I,f'} = C_{I,f+_sf'}$ Gluing two cylinders along S_I gives another cylinder with parameter $f+_sf'$
- $\widehat{C_{I,f}} = T_{I,f}$ gluing the two boundaries of the cylinder, we get a torus
- $C_{I,f+_s(1,1,0)} = C_{I,f}$ if we take $f \rightarrow f+_s(1,1,0)$ the lattice remains the same and hence it just corresponds to taking different fundamental domains of the same lattice (also preserving the boundary circle S_I)
- Given a real torus (no supersymmetric direction), i.e., $T_{I,(\tau,\bar{\tau},0)}$, we have

$$T_{g \cdot (I,\tau)} = T_{I,\tau}$$

where $g \in \text{SL}_2(\mathbb{Z})$ acts on I, τ as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot (I, \tau) = \left(I |c\tau + d|, \frac{a\tau + b}{c\tau + d} \right)$$

Form of partition function of a torus

For any $T_{I,f}$, we

$$Z[T_{I,f}] = Z[\widehat{C_{I,f}}] = \text{str } Z[C_{I,f}]$$

Since $f \in \mathbb{R}^{2|1}$, we can write $f = (z, \bar{z}, \theta)$ for $z \in \mathbb{H}$. Thus we have

$$Z[C_{I,f}] = A_I(z, \bar{z}) + \theta B_I(z, \bar{z})$$

where A_I, B_I are operators on $Z[S_I]$. Z is an even quantity as it is observable and thus since θ is odd, B must be odd too. For the same reasons A must be even.

The relation $C_{I,(z+1,\bar{z}+1,\theta)} = C_{I,(z,\bar{z},\theta)}$ gives us that

$$A_I(z+1, \bar{z}+1) = A_I(z, \bar{z})$$

$$B_I(z+1, \bar{z}+1) = B_I(z, \bar{z})$$

Additional Relations

Since $\mathbf{C}_{I,z_1,\bar{z}_1,\theta_1} \circ \mathbf{C}_{I,z_2,\bar{z}_2,\theta_2} = \mathbf{C}_{I,z_1+z_2,\bar{z}_1+\bar{z}_2+\theta_1\theta_2,\theta_1+\theta_2}$, we obtain the following equation by acting by $\mathbf{Z}$

$$\begin{aligned} & (\mathbf{A}_I(z_1, \bar{z}_1) + \theta_1 \mathbf{B}_I(z_1, \bar{z}_1)) (\mathbf{A}_I(z_2, \bar{z}_2) + \theta_2 \mathbf{B}_I(z_2, \bar{z}_2)) \\ &= \mathbf{A}_I(z_1 + z_2, \bar{z}_1 + \bar{z}_2 + \theta_1\theta_2) \\ & \quad + (\theta_1 + \theta_2) \mathbf{B}_I(z_1 + z_2, \bar{z}_1 + \bar{z}_2 + \theta_1\theta_2) \end{aligned}$$

On Taylor expanding and identifying like terms, we get the relations

$$\begin{aligned} \mathbf{A}_I(z_1, \bar{z}_1) \mathbf{A}_I(z_2, \bar{z}_2) &= \mathbf{A}_I(z_1 + z_2, \bar{z}_1 + \bar{z}_2) \\ \mathbf{A}_I(z_1, \bar{z}_1) \mathbf{B}_I(z_2, \bar{z}_2) &= \mathbf{B}_I(z_1, \bar{z}_1) \mathbf{A}_I(z_2, \bar{z}_2) = \mathbf{B}_I(z_1 + z_2, \bar{z}_1 + \bar{z}_2) \\ \bar{\partial} \mathbf{A}(z_1 + z_2, \bar{z}_1 + \bar{z}_2) &= - \mathbf{B}(z_1, \bar{z}_1) \mathbf{B}(z_2, \bar{z}_2) \end{aligned} \tag{1}$$

Suppressing the 'super' dimension

Consider a torus with no extension in the super directions, that is, a torus with $\mathbf{f} = (z, \bar{z}, 0)$. Then we define a function

$$\mathcal{Z}: \mathbb{R}^+ \times \mathbb{H} \rightarrow \mathbb{C}$$

defined by $\mathcal{Z}(l, z, \bar{z}) = \mathcal{Z} \left[\mathbf{T}_{l,(z,\bar{z},0)} \right]$ Then we have

$$\mathcal{Z}(l, z, \bar{z}) = \text{str } \mathbf{A}_l(z, \bar{z})$$

Meromorphic

$$\bar{\partial} \mathbf{A}_I(z, \bar{z}) = -\mathbf{B}_I\left(\frac{z}{2}, \frac{\bar{z}}{2}\right)^2 = -\frac{1}{2} \left\{ \mathbf{B}_I\left(\frac{z}{2}, \frac{\bar{z}}{2}\right), \mathbf{B}_I\left(\frac{z}{2}, \frac{\bar{z}}{2}\right) \right\}$$

$$\bar{\partial} \operatorname{str} \mathbf{A}_I(z, \bar{z}) = 0$$

We use the fact that str commutes with $\bar{\partial}$ and the supertrace of supercommutator vanishes.

Theorem 1 Expansion

Since $\mathbf{A}_I(z_1, \bar{z}_1) \mathbf{A}_I(z_2, \bar{z}_2) = \mathbf{A}_I(z_1 + z_2, \bar{z}_1 + \bar{z}_2)$, we see that all the $\mathbf{A}_I(z, \bar{z})$ commute (for fixed I). Thus, we diagonalise all the $\mathbf{A}_I(z, \bar{z})$ simultaneously. This will give us eigenvalues $\lambda_i^{(I)}(z, \bar{z}) \in \mathbb{C}$. Since,

$$\begin{aligned}\mathbf{A}_I(z_1, \bar{z}_1) \mathbf{A}_I(z_2, \bar{z}_2) &= \mathbf{A}_I(z_1 + z_2, \bar{z}_1 + \bar{z}_2) \\ \implies \lambda_i^{(I)}(z_1, \bar{z}_1) \lambda_i^{(I)}(z_2, \bar{z}_2) &= \lambda_i^{(I)}(z_1 + z_2, \bar{z}_1 + \bar{z}_2) \\ \implies \lambda_i^{(I)}(z_1, \bar{z}_1) &= 0 \text{ or } \lambda_i^{(I)}(z, \bar{z}) = \exp(2\pi i(a_I z - b_I \bar{z}))\end{aligned}$$

Since $\mathbf{A}_I(z + 1, \bar{z} + 1) = \mathbf{A}_I(z, \bar{z})$, we obtain that $a_I - b_I \in \mathbb{Z}$.

Theorem 1

Let $H_{a,b}(z, \bar{z}, I)$ be the eigenspace of $A_I(z, \bar{z})$ corresponding to the eigenvalue $q^a \bar{q}^b$ where $q = e^{2\pi i z}$. It will only consist of 0 if that particular pairing of a, b does not give rise to an eigenvalue of $A_I(z, \bar{z})$. Let $H_0(z, \bar{z}, I)$ be the eigenspace corresponding to the eigenvalue 0.

The superdimensions of $H_{a,b}(z, \bar{z}, I)$, $H_0(z, \bar{z}, I)$ are integers and thus, when we vary $z, \bar{z}, I$ continuously, they cannot change (as integers are a discrete space). Thus, these quantities are independent of $z, \bar{z}, I$. We denote these constant values by $h_{a,b}, h_0$.
Now,

$$\begin{aligned} \text{str } A_I(z, \bar{z}) &= \sum_{a,b} q^a \bar{q}^b \text{sdim } H_{a,b}(z, \bar{z}, I) \\ &= \sum_{a,b} q^a \bar{q}^b h_{a,b} \end{aligned}$$

Theorem 1

Since $\text{str } \mathbf{A}_l(z, \bar{z})$ is meromorphic, we obtain that $\mathbf{h}_{a,b} = \mathbf{0}$ for $b \neq 0$. Thus

$$\text{str } \mathbf{A}_l(z, \bar{z}) = \sum_{a \in \mathbb{Z}} \mathbf{q}^a \mathbf{h}_{a,0}$$

Two above two things together show that $\mathcal{Z}(l, z, \bar{z})$ is meromorphic, independent of l and has a integral fourier coefficients.

Since $\mathcal{Z}$ is meromorphic, it can only have a pole as $z \rightarrow i\infty$ ($q \rightarrow 0$). Since meromorphic functions can't have essential singularities, the sum over a needs to be bounded below.

Theorem 1

Modular invariance

Given any $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we act on I, τ as

$$I, \tau \mapsto I |c\tau + d|, \frac{a\tau + b}{c\tau + d}$$

Any two lattices linked by an elements of $\mathrm{SL}_2(\mathbb{Z})$ give rise to the same torus, thus we have

$$\mathcal{Z}\left(I, \frac{a\tau + b}{c\tau + d}\right) = \mathcal{Z}\left(I |c\tau + d|, \frac{a\tau + b}{c\tau + d}\right) = \mathcal{Z}(I, \tau)$$

First equality is because of the fact that $\mathcal{Z}$ does not depend on I , and the second one as the values of I, τ are linked by $\mathrm{SL}_2(\mathbb{Z})$, and hence are actually the same torus, so the partition function must take the same value.

Deformation invariant

A consequence of the discreteness of integers and continuity of the partition function with respect to the Riemannian structure on the torus, implies that any two SQFTs connected by a path must give rise to the same modular form. We actually have a surjection

$$\pi_0(\mathbf{SQFTs}) \rightarrow \text{Integral modular functions}$$

Any SQFT is path connected to an SCFT by RG flow, and thus we see that the map below is also surjective.

$$\mathbf{SCFTs} \rightarrow \text{Integral modular functions}$$

Sigma Models

A sigma model is a collection of QFTs (one for each manifold $\mathbf{X}$) such that the fields on a closed manifold $\mathbf{M}$ are maps $\mathbf{M} \rightarrow \mathbf{X}$.

That is, the partition function is given by

$$Z_{\mathbf{X}}[\mathbf{M}] = \int_{\psi \in \text{Maps}[\mathbf{M}, \mathbf{X}]} \mathcal{D}\psi \exp(i\mathbf{S}[\psi])$$

Now, take $\mathbf{M} = \mathbf{T}_{I,\tau}$. Then we obtain a map

$$\begin{aligned} \Phi: \text{Manifolds} &\rightarrow \text{Functions mapping } I, \tau \text{ to } \mathbb{C} \\ \mathbf{X} &\mapsto Z_{\mathbf{X}}[\mathbf{T}_{I,\tau}] \end{aligned}$$

Respects II

Now, we look at what happens on a manifold $\mathbf{X} = \mathbf{X}_1 \amalg \mathbf{X}_2$. If $\mathbf{M}$ is connected, smooth maps into $\mathbf{X}$ will either map all of $\mathbf{M}$ into $\mathbf{X}_1$ or into $\mathbf{X}_2$. So we have

$$\text{Maps}[\mathbf{M}, \mathbf{X}] \cong \text{Maps}[\mathbf{M}, \mathbf{X}_1] \amalg \text{Maps}[\mathbf{M}, \mathbf{X}_2]$$

Thus,

$$\begin{aligned} \mathbf{Z}_{\mathbf{X}_1 \amalg \mathbf{X}_2}[\mathbf{M}] &= \int_{\psi \in \text{Maps}[\mathbf{M}, \mathbf{X}]} \mathcal{D}\psi \exp(i\mathbf{S}[\psi]) \\ &= \int_{\psi \in \text{Maps}[\mathbf{M}, \mathbf{X}_1]} \mathcal{D}\psi \exp(i\mathbf{S}[\psi]) + \int_{\psi \in \text{Maps}[\mathbf{M}, \mathbf{X}_2]} \mathcal{D}\psi \exp(i\mathbf{S}[\psi]) \\ &= \mathbf{Z}_{\mathbf{X}_1}[\mathbf{M}] + \mathbf{Z}_{\mathbf{X}_2}[\mathbf{M}] \end{aligned}$$

Since $\mathbf{T}_{I,\tau}$ is connected, this implies that

$$\Phi[\mathbf{X}_1 \amalg \mathbf{X}_2] = \Phi[\mathbf{X}_1] + \Phi[\mathbf{X}_2]$$

Respects $\times$

Consider $\mathbf{X} = \mathbf{X}_1 \times \mathbf{X}_2$. Then

$$\text{Maps}[\mathbf{M}, \mathbf{X}] \cong \text{Maps}[\mathbf{M}, \mathbf{X}_1] \times \text{Maps}[\mathbf{M}, \mathbf{X}_2]$$

Given a $\psi: \mathbf{M} \rightarrow \mathbf{X}$, let ψ_1, ψ_2 be the projections on $\mathbf{X}_1, \mathbf{X}_2$.

These completely determine ψ . A sigma model is called linear if

$$\mathbf{S}[\psi_1, \psi_2] = \mathbf{S}[\psi_1] + \mathbf{S}[\psi_2]$$

For linear sigma models, we have

$$\begin{aligned} \mathbf{Z}_{\mathbf{X}_1 \times \mathbf{X}_2}[\mathbf{M}] &= \int_{\text{Maps}[\mathbf{M}, \mathbf{X}_1] \times \text{Maps}[\mathbf{M}, \mathbf{X}_2]} \mathcal{D}\psi_1 \mathcal{D}\psi_2 \exp(i\mathbf{S}[\psi_1] + i\mathbf{S}[\psi_2]) \\ &= \int_{\text{Maps}[\mathbf{M}, \mathbf{X}_1]} \mathcal{D}\psi_1 \exp(i\mathbf{S}[\psi_1]) \\ &\quad \times \int_{\text{Maps}[\mathbf{M}, \mathbf{X}_2]} \mathcal{D}\psi_2 \exp(i\mathbf{S}[\psi_2]) \\ &= \mathbf{Z}_{\mathbf{X}_1}[\mathbf{M}] \mathbf{Z}_{\mathbf{X}_2}[\mathbf{M}] \end{aligned}$$

Sigma models

Thus, taking $M = T_{I,\tau}$

$$\Phi(X_1 \times X_2) = \Phi(X_1) \Phi(X_2)$$

Respects additive identity

Take $\mathbf{X} = \emptyset$. If $\mathbf{M}$ is nonempty, then $\text{Maps}[\mathbf{M}, \emptyset] = \emptyset$. Thus, the partition function is going to be an integral over an empty set, and thus $\mathbf{0}$. Thus

$$\Phi(\emptyset) = \mathbf{0}$$

Respects multiplicative identity

Take $\mathbf{X} = \mathbf{pt}$, then $\mathbf{Maps} [\mathbf{M}, \mathbf{pt}] \cong \mathbf{pt}$. Now, $\mathbf{D}\Phi$ is supposed to represent the measure on the space of maps, but the measure on a discrete space is just the cardinality. Now, consider $\mathbf{pt} \times \mathbf{pt} \cong \mathbf{pt}$. From linearity of the sigma model, we obtain

$$\mathbf{S} [\mathbf{pt}] = \mathbf{S} [\mathbf{pt}] + \mathbf{S} [\mathbf{pt}]$$

Thus, $\mathbf{S} [\mathbf{pt}] = \mathbf{0}$. The partition function becomes

$$\mathbf{Z}_{\mathbf{pt}} [\mathbf{M}] = \int_{\mathbf{pt}} d\text{card} \exp (\mathbf{0}) = \text{card} (\mathbf{pt}) = \mathbf{1}$$

Thus

$$\Phi (\mathbf{pt}) = \mathbf{1}$$

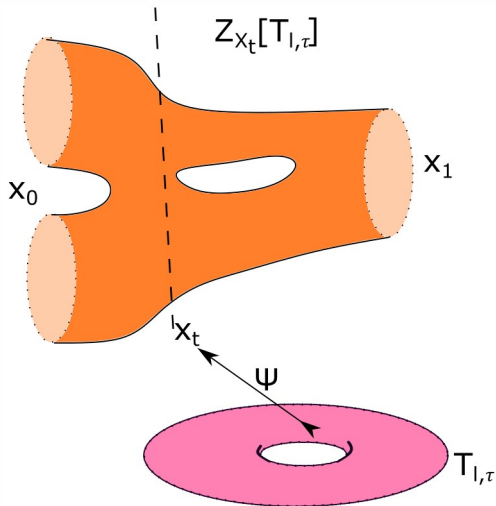
Respects the cobordism equivalence

Given two manifolds $\mathbf{X}_0, \mathbf{X}_1$, a cobordism between them Σ is a manifold such that $\partial\Sigma = \bar{\mathbf{X}}_1 \amalg \mathbf{X}_2$. Slice Σ into fibres $\mathbf{X}_t$ for each $t \in [0, 1]$ such that $\mathbf{X}_t$ match the boundaries at $t = 0, 1$. Now, consider a supersymmetric sigma model, then we obtain a map

$$\begin{aligned} p_\Sigma : [0, 1] &\rightarrow \text{Integral Modular Functions} \\ t &\mapsto Z_{\mathbf{X}_t} [T_{I,\tau}] \end{aligned}$$

Since integral modular forms are discrete (as integer coefficients cannot deform), we know that p_Σ must be a constant function. Thus, if there a cobordism from $\mathbf{X}_1, \mathbf{X}_2$, then

$$\Phi(\mathbf{X}_1) = \Phi(\mathbf{X}_2)$$



Figure

Elliptic Genera

We say two manifolds are cobordism equivalent if there is a cobordism between them. Define the cobordism ring as

$$\Omega_n = \frac{\text{set of manifolds of dimension } n}{\text{cobordism equivalence}}$$

$$\Omega = \bigoplus_{n=0}^{\infty} \Omega_n$$

Addition is given by disjoint union and multiplication by the cartesian product.

An elliptic genus is a ring homomorphism

$$\Phi: \Omega \rightarrow \text{Integral modular functions}$$

Sigma models

In the previous slides, we just showed that given any 2 dimensional $\mathcal{N} = (0, 1)$ supersymmetric linear sigma model $\mathbf{Z}$, we obtain an elliptic genus defined by

$$\mathbf{X} \mapsto \mathbf{Z}_{\mathbf{X}} [T_{I,\tau}]$$

Fun Fact: In [Och87], a universal elliptic genus was constructed. Witten [Wit87] showed that this universal elliptic genus comes from $E_8 \times E_8$ heterotic string theory in the manner described above.

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